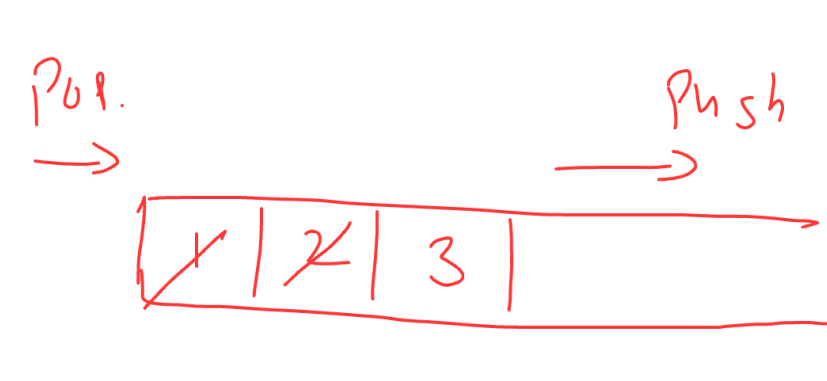
Queue.

FIFO → First In First Out

push(1)
push(2)
push(3)
pop()
pop()


Pop
~~1~~ ~~2~~ 3
Push

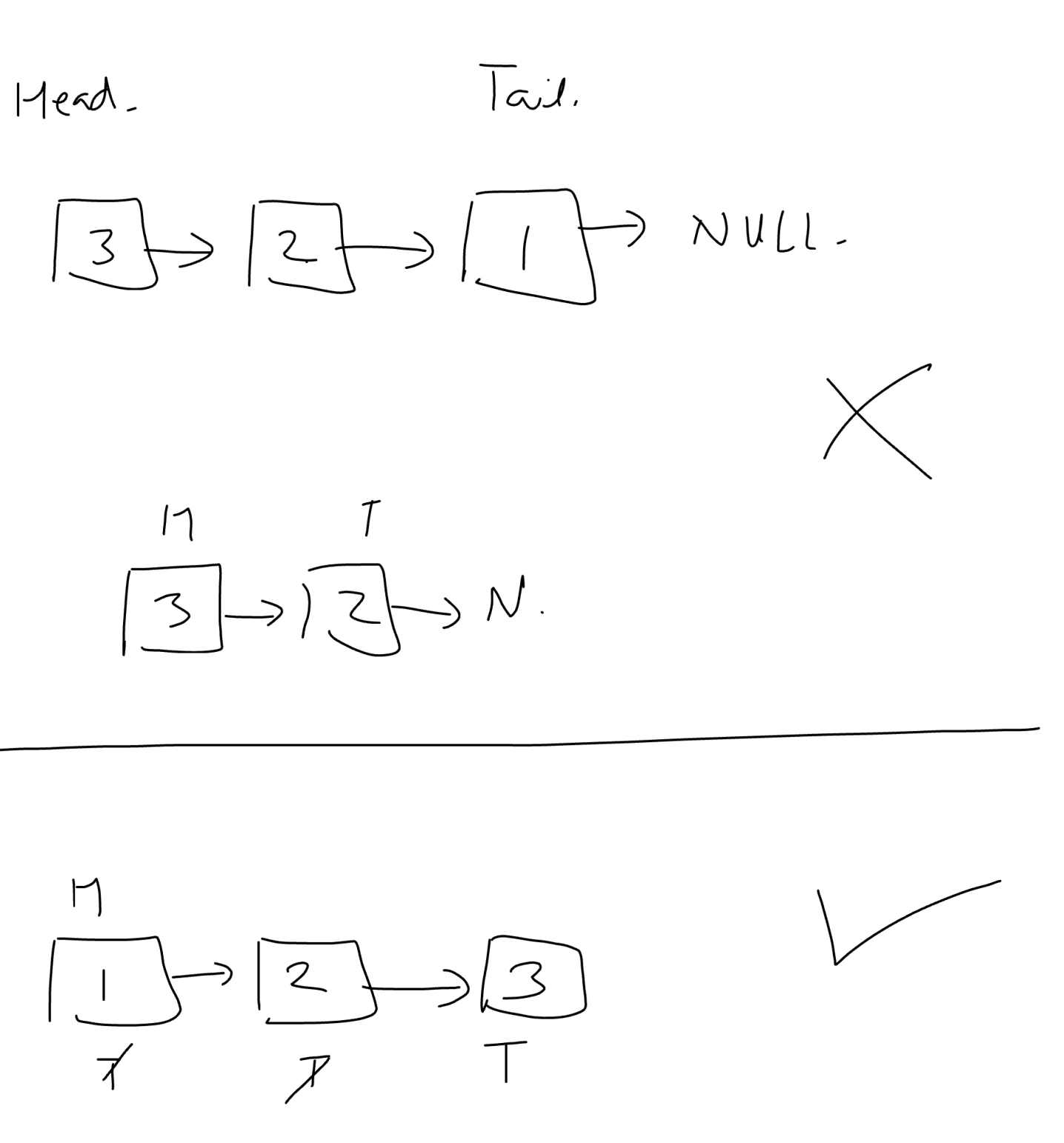
push()
pop()
front()
size()
isEmpty()

Queue using Linked list.

```
class Queue {
    private:
        LinkedList ll;
    public:
        push()
        pop()
        front()
        size()
        isEmpty()
}
```

	Queue.	L.L (with Tail ref).
O(1)	← push()	push at the end.
O(1)	← pop()	delete at the beginning.
O(1)	← front()	return head->data.

push(1)
push(2)
push(3)
→ front()
⇒ pop()
push(4)
pop()
pop()
pop()


Head.
Tail.
3
2
1
NULL.
✗
H
T
3
2
N.
H
1
2
3
T
✓

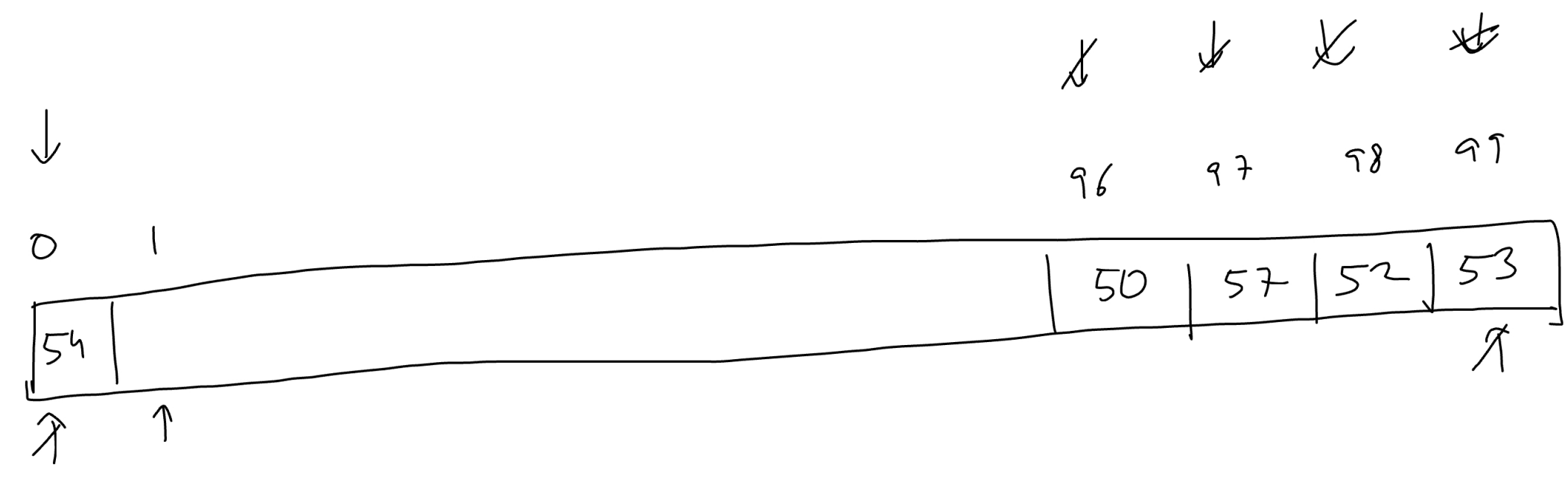
Queue Using Static Array

```
class Queue {
    private:
        int arr[100];
    public:
        .
        .
        .
        .
}
```

0
1
2
3
4
99.
1
2
3
4

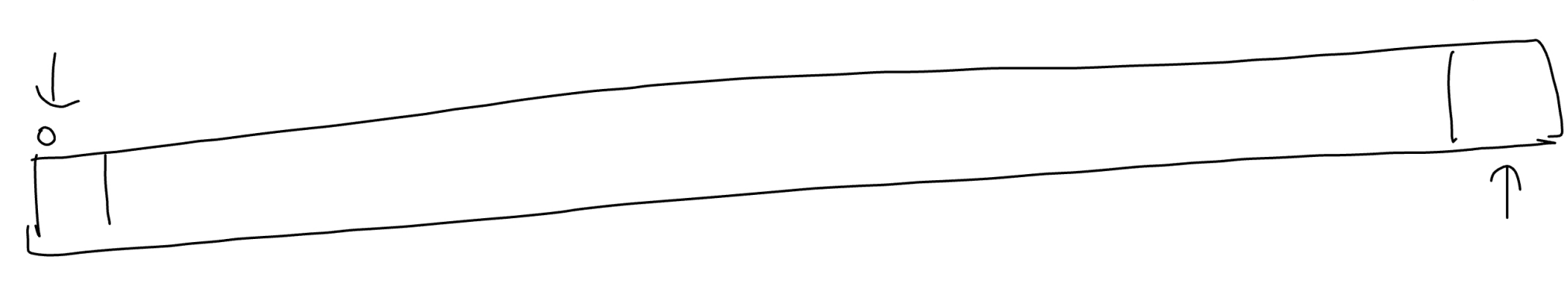
right = ~~0~~ ~~1~~ ~~2~~ 3 4

push(1)
push(2)
push(3)
→ front()
⇒ pop() ←
→ push(4) → front()
→ pop() → front()
pop() → front()
pop()


0
1
54
96
97
98
99
50
57
52
53

right = right + 1 ✗
→ right = (right + 1) % 100 ✓

left = left + 1 ✗
→ left = (left + 1) % 100 ✓



Reverse a Queue.

Queue: front → [4, 3, 1, 10, 2, 6]
O/P: [6, 2, 10, 1, 3, 4]

→ [~~4~~, ~~3~~, ~~1~~, ~~10~~, ~~2~~, ~~6~~].

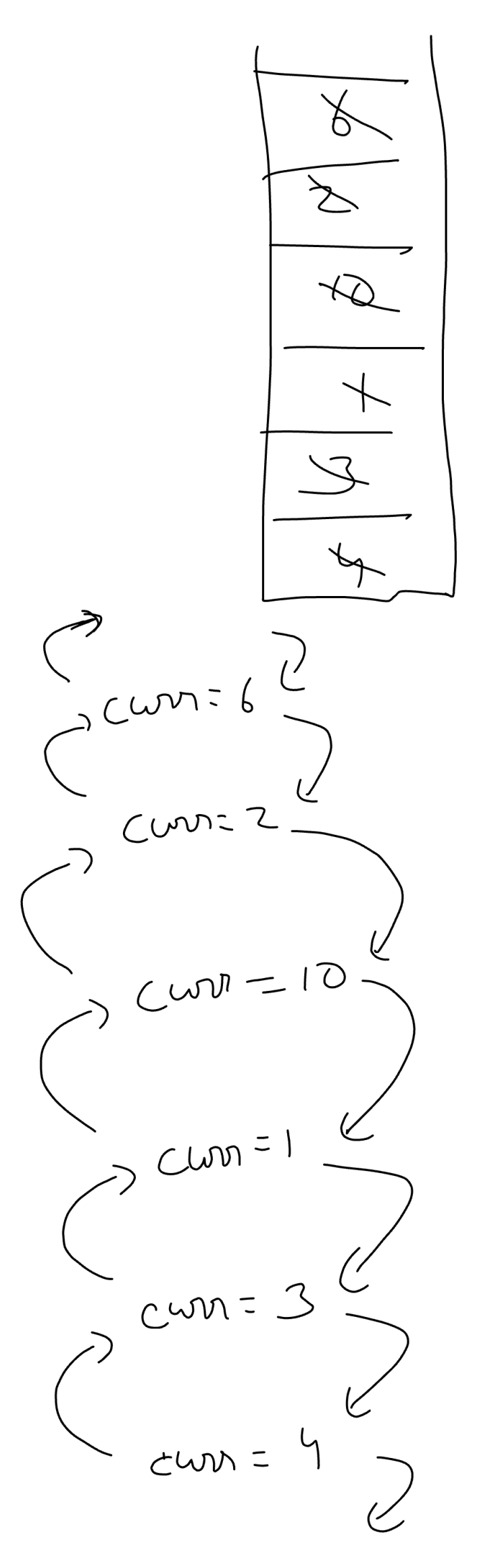
[~~4~~, ~~3~~, ~~1~~, ~~10~~, ~~2~~, ~~6~~].

[6, 2, 10, 1, 3, 4]

[~~4~~, ~~3~~, ~~1~~, ~~10~~, ~~2~~, ~~6~~].

[6, 2, 10, 1, 3, 4]

TC = O(n)
AS = O(n).


~~6~~
~~2~~
~~10~~
~~1~~
~~3~~
~~4~~
curr = 6
curr = 2
curr = 10
curr = 1
curr = 3
curr = 4

"1010"
$2^3\ 2^2\ 2^1\ 2^0$

$0 \times 2^0 + 1 \times 2^1 + 0 \times 2^2 + 1 \times 2^3 = 0 + 2 + 0 + 8 = 10.$

~~0000~~
0001
0010
0011
0100
0101
0110
0111
1000
1001
1010
1011
1100
→ 1101
1110
1111

~~0~~
~~1~~
~~10~~
~~11~~ ←
~~100~~ ←
→ 101 ←
110 ←
111 ←
1000 ←
1001 ←
1010 ←
⋮

value = 13

	13
1	6
0	3
1	1
1	0

TC = O(log n)
AS = O(1).

BF. TC = O(n log n)
AS = O(1)

n = ~~10~~ ~~8~~ 10

queue: ~~1~~, ~~10~~, ~~11~~, ~~100~~, ~~101~~, 110, 111, 1000, 1001, 1010, 1011

res: 1, 10, 11, 100, 101

TC = O(n)
AS = $O\left(\frac{n}{2} + \log n\right)$ = O(n log n).

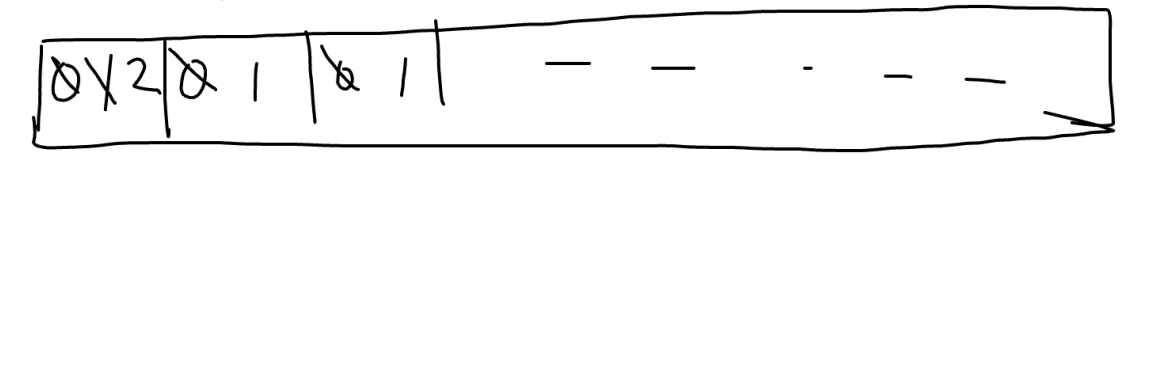
---

geeksforgeeks.

geeks . . . .

I/P: aabc
O/P: a#bb

aabc


a
b
c
0
1
2
25.
~~0~~ ~~1~~ 2
~~0~~ 1
~~0~~ 1

I/P: aacb
O/P: a#cc

aabc
a#bb

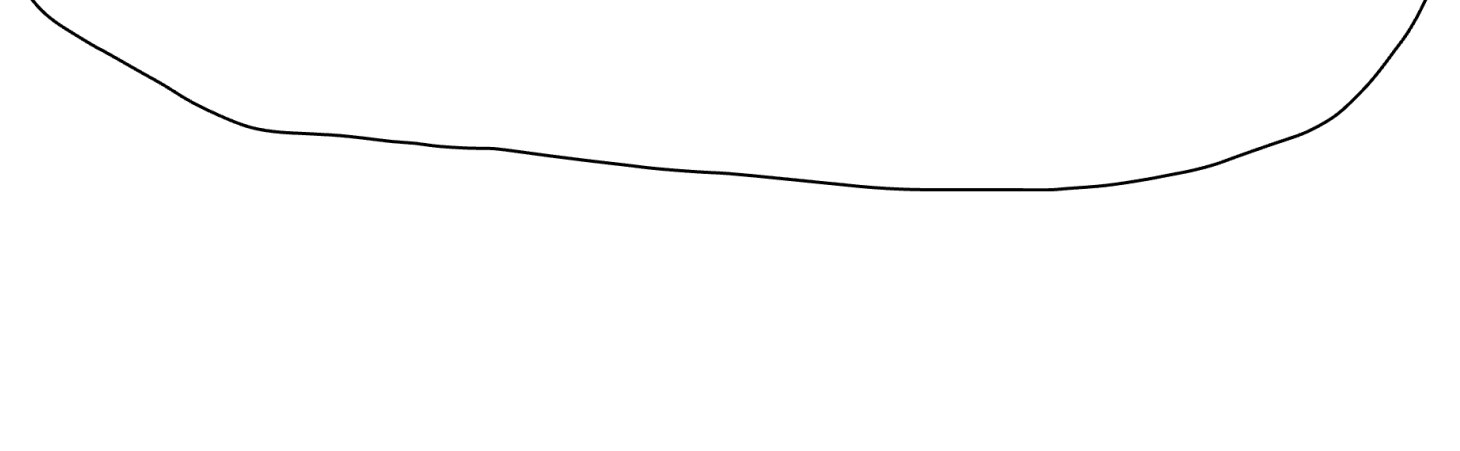
BF. TC = O(n²)
AS = O(1).

i: 0 → n−1
j: 0 → i

aabcbddcaf
a#bbc

a → ~~1~~ 2
b → ~~1~~ 2
c → 1

aabcbddcaf.
a#bbccc##f

a → ~~1~~ ~~2~~ 3
b → ~~1~~ 2
c → ~~1~~ 2
d → ~~1~~ 2
f → 1